

Case Report

Real-time unmet demand-driven relocation policy to improve service capacity of shared E-mopeds

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ABSTRACT

The growing popularity of shared electric moped scooters (E-mopeds) as a sustainable and convenient transport option faces a fundamental challenge: demand often exceeds supply, necessitating efficient real-time relocation strategies. To address this, we propose an unmet demand-driven relocation policy to improve service efficiency and E-moped utilization. This policy leverages recurrent and diffusion convolutional graph neural networks alongside linear programming. Regional E-moped relocation is suggested to be triggered when its demand falls below a threshold parameter. We demonstrate that the unmet demand strategy consistently outperforms the traditional pick-up method. At a threshold of 3, the unmet demand approach results in 286 relocated E-mopeds compared to 100 using the pick-up strategy, emphasizing the importance of our study. These findings assist operators in implementing more efficient strategies and inform policymakers in refining the maximum fleet sizes and parking space allocation.

1. Introduction

In many cities, motorcycles are the means of transportation that most people rely on. The pervasive use of motorcycles (or scooters) has increased substantial traffic and environmental concerns, as highlighted in previous studies (Chalya et al., 2010; Hsu et al., 2003; Hussain et al., 2005; Kitamura et al., 2018). In response to this issue, the government has endorsed the adoption of electric moped scooters (E-mopeds) and shared E-moped services, particularly in major cities like Taipei, leading to noteworthy success. The E-moped, a two-wheeled vehicle with a footrest platform and seat, distinguishes itself from the light stand-up E-scooter. Primarily employed for short-distance commuting, recreational use, and facilitating first-and-last-mile connectivity to public transportation, the shared E-moped's utility is heavily influenced by the urban dynamics of the origin-destination (OD) areas (Aguilera-García et al., 2020). Accessible through mobile applications, the on-demand services enable users to locate nearby available E-mopeds, with payment structured on trip duration. Service management entails periodic maintenance tasks such as battery swapping and relocation. Despite its burgeoning popularity, the emerging shared E-moped service grapples with the notable challenge of insufficient supply relative to high demand. Real-time relocation emerges as an effective strategy for addressing this issue by redistributing idle E-mopeds to regions experiencing heightened demand.

This research focuses on the real-time relocation policy driven by the unmet demand from users to enhance overall service capacity. We address a research gap in previous studies, where the common practice of using pick-up trips as relocation targets is inherently flawed. Pick-up trips (also known as pick-up demands or satisfied demands) and unmet demands are mutually exclusive components

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that constitute the overall user demand. The satisfied demand refers to successfully fulfilled customer rentals, while the unmet demand reflects dissatisfaction due to the unavailability of nearby E-mopeds (Ham et al., 2021). Pick-up demand-driven policy poses a risk to the success of real-time relocation, particularly in situations of demand and supply imbalance (Li, Cao, et al., 2018). For instance, most regions exhibit high rental frequencies during peak usage periods. The pick-up demand prediction cannot differentiate between regions that already possess sufficient supply and those that still exhibit high levels of unmet demand. Consequently, this may lead to relocation driven by pick-up inaccurately moving vehicles to areas with low unmet demand, failing to optimize overall service capacity. To achieve effective and precise relocation, it is imperative to accurately determine real-time unmet demand in each region.

In this research, we will discuss the related work in Section 2, explain the methodology in Section 3, present experimental results and comparisons in Section 4, discuss findings in Section 5, and conclude in Section 6. The main contributions of this research are listed below:

- 1) We propose a comprehensive research framework to prove the importance of unmet demand-driven policy. This framework involves demand prediction and the simulation of regional E-moped scenarios to assess each contribution to service capacity, supported by experimental results. We additionally propose a threshold parameter to determine that if the unmet demand in a region falls below the designated threshold, its E-mopeds should be dispatched to other regions.
- 2) As the accuracy of real-time unmet demand influences relocation effectiveness, we utilize a spatio-temporal graph neural network model for prediction. The model's performance and contributions from additional learning features are demonstrated and compared with the pick-up demand.

2. Related work

To our best knowledge, very few studies have discussed the unmet need, and there has been no survey of the unmet demand-driven policy for real-time shared E-moped relocation. We will briefly review the relevant research for similar shared mobilities, including bikes and E-scooters, as their on-demand services share common characteristics, such as providing short-distance trips, integrating public transportation networks, and connecting with urban functionalities. Pal and Zhang (2017) proposed a mixed integer linear program method for solving the static complete relocation problem, which only rebalanced the fleets at night and did not consider the dynamic demand. Jiuhe et al. (2021) tackled the shared E-moped relocation problem by defining various relocation zones with a clustering algorithm and subsequently optimizing routing through the genetic algorithm (GA) (Holland, 1992); however, they randomly generated the demand and did not discuss the temporal unmet demand issue. Kim et al. (2023) also utilized the GA and adopted the historical trip-shared E-scooter data in Korea. They addressed the supply-demand imbalance by considering an assumed proportion of the potential demand, yet lacked a specific discussion on the unmet demand. Caggiani et al. (2018) built a framework for shared dockless bikes, including the pick-up demand prediction based on a recurrent neural network (RNN) model (Rumelhart et al., 1985) and dynamic relocation strategies optimized by the integration with the GA and other heuristic algorithms. In addition, more recent studies have started to use deep reinforcement learning (DRL) methods (Lillicrap et al., 2015) to address more complex real-time relocation problems. Pan et al. (2019) proposed a hierarchical reinforcement pricing algorithm integrated with DRL and RNN models to incentivize users to rebalance shared dockless bikes with the historical trip data in China. However, they did not specify the unmet demand. Li, Zheng, and Yang (2018) implemented a Deep-Q-Network (DQN) (Mnih et al., 2015) integrated with the RNN prediction model, tackling the shared docked bike relocation problem in the U.S.A. with historical pick-up and drop-off trips. They stated the importance of weather, events, and the correlation between bike stations affecting the trip demand. Losapio et al. (2021) also utilized the DQN to simultaneously consider the integrated strategies of user incentive, battery swapping, and relocation. A synthetic demand simulator was utilized, including the pick-up and unmet demand based on the historical shared E-scooter data (Ciociola et al., 2020).

On the other hand, unlike most studies (He & Shin, 2020; Kim et al., 2022; Phithakkitnukoon et al., 2021; Xu et al., 2021) utilizing the pick-up as the demand target, the research on the shared E-scooter demand prediction (Ham et al., 2021) notably indicated the essence of the unmet demand for the relocation strategy and maximization of the user utility. It proposed an algorithm to collect and define the unmet demand according to feedback from mobile app usage. If a user zoomed in on the map near their present position within a 250-m distance, it was defined as a user search record. And if the user failed to rent one in the following 15 minutes successfully, it would also be classified as unmet demand. Although this research effectively used map zoom-in data to define user search and unmet demand, it did not specifically discuss how to apply the unmet demand to the relocation and needed more verification of its superiority compared to the pick-up demand. Based on this research, we collect the user search data through the shared service app and the trip data through E-moped's global positioning system (GPS) feedback to define the unmet demand. Summarizing the current approaches regarding shared E-moped relocation, they utilized pick-up trips as relocation targets. Some regions with high unmet demand might be overlooked if historical data revealed relatively low trip numbers there. Therefore, we aim to solve this research gap by proposing a real-world and real-time shared E-moped relocation policy driven by the unmet demand. This policy is verified with the numerical results to improve the overall service capacity compared to the pick-up demand in the following sections.

3. Methodology

3.1. Research framework

Overall, we propose a research framework for the unmet demand-driven relocation policy in Fig. 1, summarizing as follows.

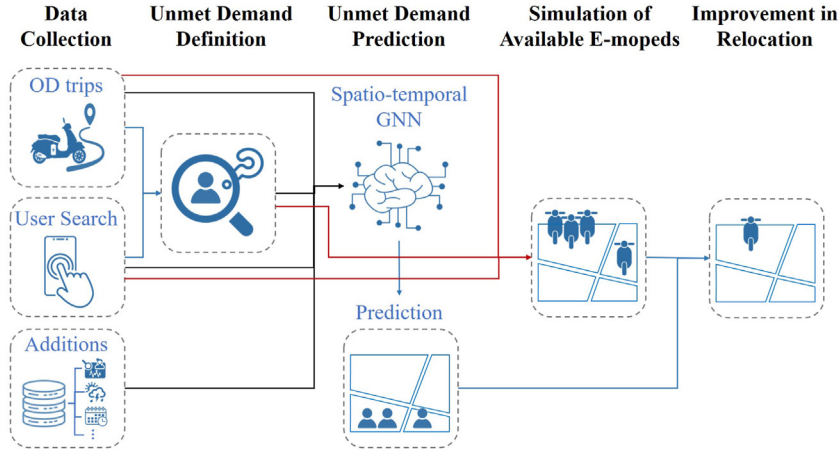


Fig. 1. The research framework of shared E-moped search demand prediction and evaluation of potential improvement in relocation.

- 1) *Data collection.* The most crucial data regarding our relocation research is user search records and E-moped OD trips. We additionally collect the built environment and weather condition data to assist in understanding trip behaviors and improve relocation accuracy. The collection, definition, preprocessing, and descriptive statistics of the data are introduced in Section 3.3.
- 2) *Unmet demand definition.* We utilize the difference between user search and OD-trip data to define the unmet demand as the relocation target, depicted in Section 3.2.
- 3) *Unmet demand prediction.* We utilize the spatio-temporal graph neural network model proposed by Song et al. (2023) to predict the unmet demand of shared E-mopeds described in Section 3.4, and its performance is revealed in Supplementary Materials 3.
- 4) *Simulation of available E-mopeds.* To execute our relocation policy, in addition to the information on pick-up and drop-off trips, the historical regional status of the available E-mopeds must be acquired. As this information is challenging to obtain, we propose a feasible alternative simulation approach in Section 3.5.
- 5) *Improvement in relocation.* Relocation effectiveness is based on increasing service capacity while maintaining a certain level of service cost and a fixed fleet of E-mopeds. To accomplish this, we determine an appropriate level of additional relocation driven by the unmet demand predictions. A method in Section 3.5 evaluates the potential increase in demand satisfaction from the additional relocation as an indicator of improvement. The experimental results demonstrate the comparison between the unmet and pick-up demand-driven policies in Section 4.2.

3.2. Problem definition

This research defines unmet demand as the number of users who intend to rent a shared E-moped but fail to find one nearby. An available E-moped can be identified if an E-moped is available to be rented. The quantities of unmet demands and available E-mopeds are further accumulated and mapped into each target region (defined in Section 3.3.) at different periods. Based on the above definitions, we introduce the concept of real-time relocation driven by the unmet demand and its improvement in service capacity with an illustrative example in Fig. 2. There are 4 available E-mopeds for rentals in region $i = 0$. The real-time predictions for the unmet demand with 2, 2, and 1 E-moped in the region $i = 1$, $i = 2$, and $i = 3$, respectively, are shown in Fig. 2a. Next, based on the demand predictions and available E-mopeds, the real-time relocation strategy in Fig. 2b instructs the transport truck to pick up 4 E-mopeds from region $i = 0$ and place 2 E-mopeds in each $i = 1$ and $i = 2$. These two regions are assigned because of their higher unmet demand than region $i = 3$. Afterward, Fig. 2c reveals that the overall unmet demand drops from 5 to 1 E-moped, i.e., this strategy improves the service capacity with 4 E-mopeds.

As illustrated in the above example, the priority is to obtain the real-time unmet demand to achieve relocation effectiveness. Due to the difficulty in directly measuring unmet demands, we estimate it using the user search and trip data sets provided by the shared service operator. Each user search record is transmitted and then stored in the service central system if a user opens the service app and taps on the mobile screen to find available E-mopeds within 250 m of his/her current location (Ham et al., 2021). Each trip data with OD information, along with the corresponding times, is recorded by sensors on each E-moped if it is rented. Part of trip data, pick-up demand, also called satisfied demand, represents the number of E-mopeds that fulfill the user's rental demand. These two data sets are further mapped and accumulated into target regions at different periods. Ignoring minor errors caused by human uncertainties (such as unexpected changes in the intention of rental) and data defects (such as GPS drifting and sensor errors), two data sets should be theoretically matched, and the difference between the number of user searches and pick-up demands can represent the number of unmet demands in the same region in the same period. Thus, we define unmet demand as the search demand minus the satisfied demand, illustrated in Eq. (1):

$$d_{i,t}^{\text{unmet}} = d_{i,t}^{\text{search}} - d_{i,t}^{\text{pick}} \geq 0, \quad \forall i \in \mathbf{I}_{\text{region}}, \forall t \in \mathbf{T} \quad (1)$$

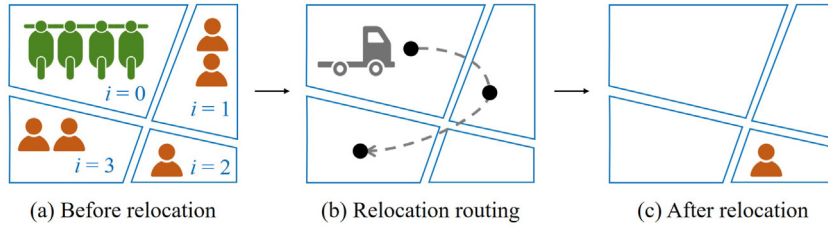


Fig. 2. The concept of relocation is moving the available E-mopeds to high-demanding regions.

where $d_{i,t}^{unmet}$, $d_{i,t}^{search}$, and $d_{i,t}^{pick}$ respectively represent the unmet demand, user search demand, and pick-up demand in region $i \in I_{region}$ in period $t \in T$. I_{region} and T denote a set of overall regions and periods in this research, while N_{region} and N_t denote each set's length (total number of units), respectively. Each demand can be quantified by the number of E-mopeds. In this research, we define demand prediction ($\hat{d}_{i,t}^{unmet}$) as our relocation target, which is obtained from the prediction model. This model minimizes the loss between the unmet demand's ground truth ($d_{i,t}^{unmet}$) and prediction ($\hat{d}_{i,t}^{unmet}$), discussed in Section 3.4.

Next, we construct a principle as Eq. (2) governing the equilibrium of E-moped quantity variation for each region from period $t - 1$ to period t .

$$n_{i,t-1}^{avail} + d_{i,t}^{drop} - d_{i,t}^{pick} + n_{i,t}^{reloc} = n_{i,t}^{avail} \geq 0, \quad \forall i \in I_{region}, \forall t \in T \setminus \{0\} \quad (2)$$

where $n_{i,t-1}^{avail}$ and $n_{i,t}^{avail}$ respectively represent the number of available E-mopeds at the end of the period $t - 1$ and t , which are available to be rented in the next period (t and $t + 1$). $d_{i,t}^{pick}$ represents the drop-off demand, the number of returned E-mopeds in region i in period t . $n_{i,t}^{reloc}$ represents the number of dispatched E-mopeds already executed by the service operator in region i in period t , which we define as "existing relocation." Notice that, unlike $n_{i,t-1}^{avail}$, $n_{i,t}^{avail}$, $d_{i,t}^{pick}$, and $d_{i,t}^{drop}$ are non-negative, $n_{i,t}^{reloc}$ can be either a positive, zero, or negative value. The positive relocation ($n_{i,t}^{reloc+}$) indicates E-mopeds being placed in region i , whereas the negative relocation ($n_{i,t}^{reloc-}$) denotes E-mopeds moving away from region i . Given Eq. (2), we can derive Eq. (3):

$$n_{i,t-1}^{avail} + n_{i,t}^{reloc} \geq -(d_{i,t}^{drop} - d_{i,t}^{pick}), \quad \forall i \in I_{region}, \forall t \in T \setminus \{0\} \quad (3)$$

In Eq. (3), $d_{i,t}^{drop} - d_{i,t}^{pick}$, the right-hand side can be obtained from the historical trip data, while the left-hand side components $n_{i,t-1}^{avail}$ and $n_{i,t}^{reloc}$ are unknown and can be simulated based on the right-hand side in Section 3.5.

Once $n_{i,t-1}^{avail}$ and $n_{i,t}^{reloc}$ have been simulated in each region over periods, we can subsequently simulate how many more E-mopeds can be additionally relocated to reduce unmet demands by rewriting Eq. (2) as 4. Here, the additional relocation refers to the number of E-mopeds driven by the unmet demand relocation policy in contrast to the existing relocation, aiming to further enhance the overall service capability.

$$n_{i,t-1}^{avail} + d_{i,t}^{drop} - d_{i,t}^{pick} + n_{i,t}^{reloc} + \hat{n}_{i,t}^{reloc} = n_{i,t}^{avail} \geq 0, \quad \forall i \in I_{region}, \forall t \in T \setminus \{0\} \quad (4)$$

where $\hat{n}_{i,t}^{reloc}$ represents the additional relocation number of E-mopeds in region i in period t . To reduce unmet demands based on the prediction ($\hat{d}_{i,t}^{unmet}$), the additional relocation should follow the principle of transferring idle E-mopeds from the high-availability regions to high-demand regions in each period. Therefore, we can define the improvement of overall service capacity (Δ_t) as the reutilization of idle E-mopeds contributed by additional relocation, i.e., the sum of the differences between the ground truth of unmet demand ($d_{i,t}^{unmet}$) and unmet demand after additional relocation ($\hat{d}_{i,t}^{unmet}$) across all regions in each period t . It is also equivalent to the sum of additional positive relocation ($\hat{n}_{i,t}^{reloc+}$). In the experiments, we utilized the test dataset with period $t \in T_{test}$ ($T_{test} \subset T$) to evaluate Δ_t ; hence, the equation can be written in Eq. (5). We will also evaluate the total improvement (Δ_{total}) over T_{test} and average improvement per period (Δ_{avg}) in Section 4.2.

$$\Delta_t \equiv - \sum_{i=0}^{N_{region}-1} \{\hat{d}_{i,t}^{unmet} - d_{i,t}^{unmet}\} = \sum_{i=0}^{N_{region}-1} \{\hat{n}_{i,t}^{reloc} \mid \hat{n}_{i,t}^{reloc} > 0\}, \quad \forall t \in T_{test} \quad (5)$$

3.3. Data description and descriptive statistics

The data used in this research in association with its geographic information are described below:

- **Service area and target regions.** In Fig. 3, the service scope of data we have adopted is over 150 km², defined by the service operator, including most of Taipei City and parts of the New Taipei City area, briefly called Greater Taipei. To effectively predict demands for a practical relocation purpose, we split the service area based on the Basic Statistical Areas (BSAs) into the heterogeneous regions of prediction targets. BSAs (Ministry of the Interior, Taiwan, 2008) are utilized to establish a comprehensive and enduring set of fundamental geographic data. It can serve general statistical purposes and facilitate the integration of various government registrations. Compared to divisions based on grids, the BSA divisions consistently and effectively embed the socioeconomic and demographic data, contributing to feature extractions in the prediction model. Based on the BSAs, a clustering method to define the regions of prediction targets is explained in Fig. 4. Fig. 4a shows an overall BSA configuration divided by black lines within the service area. Typical machine learning clustering algorithms likely obtain the tattered and varying-size geometries, which are impractical for the service operation. We instead take advantage of the Voronoi diagram to split into the convex polygons based on the administration districts. For each district within the service area, the number of subdivisions would be based on its area. The total number of 151 divisions to restrict each one that shares an approximately identical area of 1 km² in Fig. 4b. Underlying this Voronoi configuration, the BSAs (divided by black lines) can finally be clustered as the 151 regions of prediction targets shown in different colors in Fig. 4c.
- **User search and OD-trip data.** The OD-trip data within the service area includes over 0.5 million trips in April 2021, and each trip contains the time and location information from origins to destinations. In addition, the user search data is separately collected, including the records of each customer's location and time when using the service app. Corresponding to the periods of trip data, the search data contains over 0.8 million records. Fig. 5 demonstrates the average daily user search, pick-up, and unmet demands. It can be seen that the central area experiences the highest search demands with dark green. In contrast, the regions with the highest pick-up demand for dark blue are more concentrated (around Taipei Main Station, Zhongshan, Huashan Creative Park, Xingtian Temple, Taipei Arena, Eastern District, and Tonghua Night Market). The correlation between pick-up and search demand is not simply positive. As a result, the regions with higher unmet demand with deeper red are around Taipei Main Station, Ximending, Zhongshan, National Taiwan University Hospital, Daan, Yonghe, Eastern District, Shih Hsin University, Shilin, and Neihu Technology Park. This difference between the pick-up and unmet demand also indicates that the pick-up demand is unsuitable to replace the unmet demand as the relocation target.

We further conduct a more in-depth comparison of periodic behaviors between the demands in Fig. 6. Based on the average hourly distribution, Fig. 6a reveals that the overall demand exhibits the largest degree of variation, followed by the satisfied demand, and the unmet demand shows the smallest. To reflect the service reallocation strategies (six shifts per day), we recluster each 4-h data into a period in Fig. 6b; its demand pattern is generally consistent with the hourly demands in Fig. 6a. The following experiments will also



Fig. 3. The service area in Greater Taipei is defined by the shared E-moped operator.

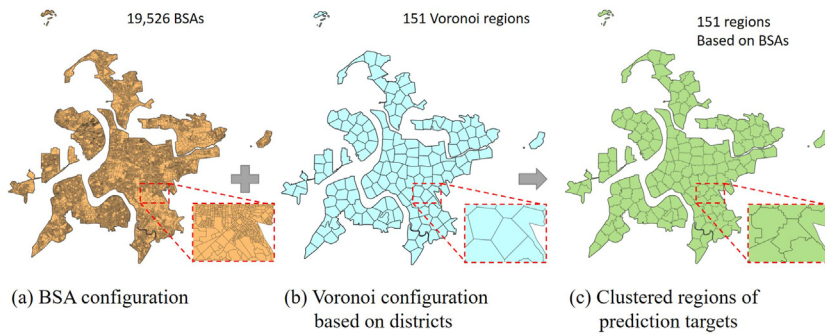


Fig. 4. The process to define the division regions of prediction targets based on BSAs.

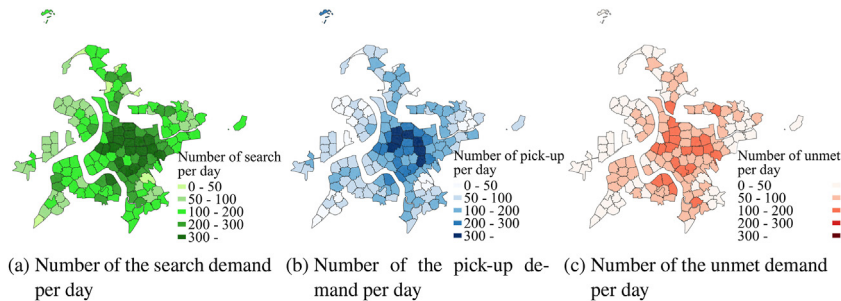


Fig. 5. The daily average number of user searches, pick-ups, and unmet demands for each region.

employ this 4-hour period approach for conducting relocation policies. On the other hand, we demonstrate three distinctive demand spatial distributions over regions in Fig. 6c. It can be seen that the unmet demand is relatively concentrated compared to others; however, the imbalance between regions is still significant. It experiences the greatest variation between 4 p.m. and 8 p.m., which differs from the search and pick-up demand between 8 p.m. and 12 a.m.

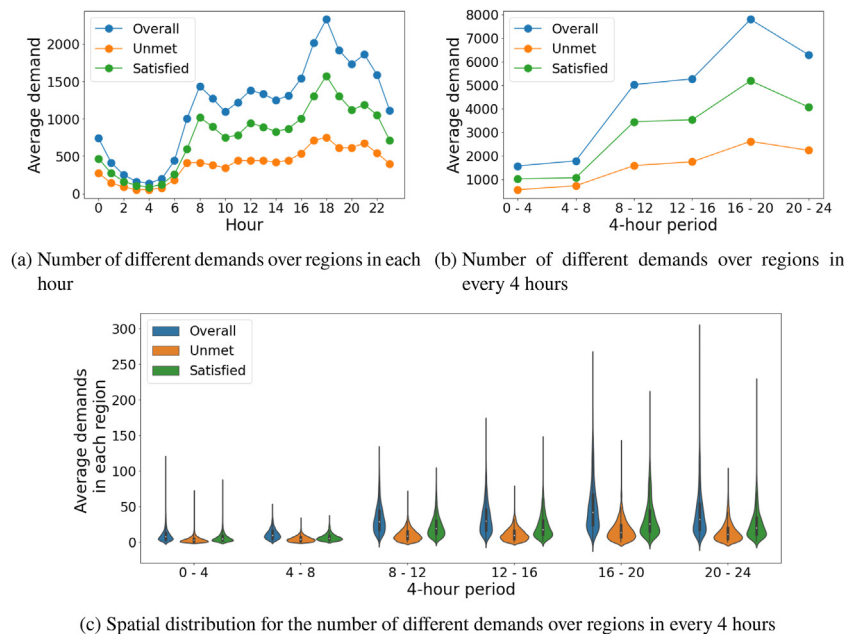


Fig. 6. Analysis of the spatial and temporal distribution between three different demands.

Built environment, weather conditions, and periodic data. Based on prior research (Li et al., 2023; Song et al., 2023; Yang et al., 2023), this research collects several data sets with essential aspects to improve the accuracy of unmet demand prediction. First, regarding the built environment, POIs (points of interest, reflecting detailed and current land uses), spatial relationships (such as street density and street intersection density), and socio-demographics (such as sex, age, and education levels) are included. Greater Taipei possesses the characteristics of a dense population, complex public transportation systems, and highly mixed land uses. Since shared E-moped usage is based on activities of daily living such as commuting, dining, shopping, and entertainment, the above factors profoundly influence trip behavior and unmet demand. To consider shared E-moped service's connection to public transportation for first-and-last-mile integration, this model also collects all public transportation location information (such as shared bikes, buses, and trains) to extract spatial relationships between them and unmet demand. Second, weather conditions also significantly impact the usage of shared E-mopeds; for instance, severe precipitation significantly reduces the user's rental intention. We collect detailed hourly average information, including temperature, precipitation, relative humidity, and wind speed from the weather stations. Third, periodic activities and special events impact the usage of shared E-mopeds. For instance, commuting activities likely lead to high usage demand in office and commercial areas during peak hours. During special holidays, areas hosting large-scale events may also experience a significant concentration of usage. We leverage numeric values representing different time intervals—hours, weekdays, months, and holidays—in the graph representations to reflect the usage demand behavior during specific periods. (See Supplementary Materials 1 for a more comprehensive data description.)

3.4. Prediction model for unmet demand

Graph neural networks (GNNs) have proven effective in many application fields (Chou et al., 2024; Jean et al., 2024; Kuo et al., 2024). To predict the real-time demand for shared micro-mobility, recent research (Ham et al., 2021; He & Shin, 2020; Kim et al., 2022; Xu et al., 2021) have focused on approaches of spatio-temporal GNNs, which revealed the superior performance compared with the classical machine learning and regression models. Leveraging the advantages of RNN, models were capable of capturing recent historical demand patterns as temporal features. The graph convolutional network (GCN) (Kipf & Welling, 2016) additionally enhanced feature representations for complex spatial relationships. Among those approaches, SpDCGRU (Song et al., 2023) achieved an overwhelming performance. It leveraged advanced diffusion convolutional layers (Li et al., 2017) to improve the message-passing of the connections between origins and destinations with comprehensive and sophisticated additional learning features. Based on this research, we modify the model to be adapted to the shared E-moped domain. Precisely, we utilize graph universal features to describe periodic factors, including days, hours, and identification of weekends and holidays. On the other hand, the node features represent the regional information that could be categorized into time-varying and static. The time-varying node features include the user search demands, pick-up demands, drop-off demands, unmet demands, and weather conditions. The static one represents the regional urban environments, including the socio-demographics, spatial relationships, POIs, and public transportation locations. Note that although parts of node feature messages are static, they are passed through dynamic edge relations to update each node state dynamically over periods. Edge features represent relationships between OD regions, including distance, geographical contiguity, and number of trips during each period. It should be noted that this research adopts the above factors as numerical vectors instead of manually simplified similarity and correlation extraction (He & Shin, 2020; Xu et al., 2021). Our sophisticated model can more directly encode the vector features with numerical values, thereby obtaining more accurate predictions. In our analysis, all of the above features significantly influence the accuracy of unmet demand prediction. The impact of periodic and weather-related factors emerged as particularly essential, ranking as the primary and secondary contributors, respectively. Detailed statistical results are documented in Supplementary Materials 3.

We utilize the mean absolute error (MAE) (Hyndman & Koehler, 2006) as the main loss function and evaluation metric, which can be defined as Eq. (6):

$$MAE = \frac{1}{N_{batch}N_{region}} \sum_{t=t}^{t+N_{batch}-1} \sum_{i=0}^{N_{region}-1} |d_{i,t}^{unmet} - \tilde{d}_{i,t}^{unmet}| \quad (6)$$

where $d_{i,t}^{unmet}$ and $\tilde{d}_{i,t}^{unmet}$ specify the ground truth and prediction in region i in period t , respectively; $|\cdot|$ is the absolute operator; N_{batch} represents the batch size, number of periods considered if in a batch training, while N_{region} denotes the sum of the evaluated periods if in the evaluation metric. MAE also indicates the difference in the number of E-mopeds. In addition, as the variations in the number of E-mopeds differ significantly across regions and periods, we also utilize the mean absolute percentage error (MAPE) (Flores, 1986) to evaluate the prediction model's performance. To modify MAPE not greatly affected by the small values, the advanced MAPE₁₀ was proposed (Tang et al., 2021; Xu et al., 2021). We consider MAPE₁₀ and MAPE₄₀ as the evaluation indicators of the regions with unmet demands larger than 10 and 40, respectively, defined as Eq. (7):

$$MAPE_n = \frac{1}{N_{batch}N_{region}} \sum_{t=t}^{t+N_{batch}-1} \sum_{i=0}^{N_{region}-1} \frac{|d_{i,t}^{unmet} - \tilde{d}_{i,t}^{unmet}|}{d_{i,t}^{unmet}}, \forall d_{i,t}^{unmet} \geq N_n \quad (7)$$

where N_n is set to 10 and 40 corresponding to MAPE₁₀ and MAPE₄₀. Here, it should be noted that N_{batch} and N_{region} only consider the number of $d_{i,t}^{unmet} \geq N_n$.

3.5. Simulation of regional E-moped situation and evaluation on demand prediction's improvement in relocation

To effectively evaluate the potential service capacity improvements from additional relocation, it is essential to understand the historical situation of the variation in the number of E-mopeds in each region over periods. Based on practical experience in the scope of greater Taipei, a worker with a truck can generally complete 25-E-moped relocation on average in 4 hours, including the time costs of transportation, placement, removal, and battery swapping. Accordingly, we define relocation capacity (cap) to describe and include the above overall relocation operation. For instance, the relocation capacity set to 150 (E-mopeds) implies that the service operator utilizes a maximum of 6 trucks to complete 150-E-moped relocation tasks during each period. The relocation capacity is a parameter for different relocation scenarios and is further divided into existing relocation capacity (cap_{exist}) and additional relocation capacity (cap_{add}). The existing relocation capacity (cap_{exist}) indicates the number of E-mopeds for the existing relocation (which has already been implemented and has an underlying relationship with E-moped trips.) As obtaining data on existing relocation ($n_{i,t}^{reloc}$) and available E-mopeds ($n_{i,t}^{avail}$) is challenging, we propose an alternative simulation method corresponding to the historical trip data. (See details in Supplementary Materials 2.)

The additional cap_{add} , which is relatively concerning in this research, represents the maximum number of E-mopeds relocated by additional trucks. This parameter varying in different scenarios reflects the additional operating costs that the service operator would like to afford at the most. This parameter is more essential because it explicitly affects the additional relocation performance that could be improved. We further propose a linear program model to simulate the additional relocation ($\hat{n}_{i,t}^{reloc}$). The upper limit for the additional relocation should be constrained by cap_{add} , $n_{i,t}^{avail}$, $d_{i,t}^{unmet}$, and $\tilde{d}_{i,t}^{unmet}$. We presume that cap_{exist} is greater than cap_{add} because of the constraint imposed by operating costs. In addition, $n_{i,t}^{avail}$ indicates the number of available E-mopeds that region i can provide in period t . To ensure the constraints in Eq. (4) after additional relocation in each period $t \in T_{test}$, we define the set of the dynamic capacity of available E-mopeds for all regions (cap_i^{avail}) by Eq. (8):

$$cap_i^{avail} = \{\min(\{n_{i,t}^{avail} | t \in T_{test}\}) | i \in I_{region}\} \quad (8)$$

where $\min(\cdot)$ represents the minimum operator along the period. cap_i^{avail} will be decreasingly updated once region j has executed negative additional relocation by Eq. (9):

$$cap_i^{avail}[j] = cap_i^{avail}[j] + \hat{n}_{i=j,t}^{reloc} \quad (9)$$

Another essential thing is that the additional relocation should target the unmet demands. In an ideal scenario, we can utilize the identification of zero unmet demands (based on the prediction $\tilde{d}_{i,t}^{unmet}$) to differentiate the regions executing positive and negative additional relocation. Note that because the total demand significantly exceeds the total supply in practice, the number of regions with zero demand in a period is significantly fewer. In addition, to ensure the customers in high-demand regions can be better satisfied, the service operators would practically give them higher priority; in other words, regions with relatively lower demand can be appropriately sacrificed. As a result, τ indicates a parameter for the threshold value of the low demand, which determines positive or negative relocation. When the unmet demand in a region is less than or equal to the value of τ , available E-mopeds will be dispatched to other high-demand regions to enhance overall service capacity. On the other hand, as demand predictions are inevitably subject to errors, this τ can be used to adjust the relocation strategy to enhance overall performance (discussed in Section 4.3). Afterward, we define the upper limits for potential satisfaction of the unmet demand ($n_{i,t}^{unmet}$), which is the minimum value between $\tilde{d}_{i,t}^{unmet}$ and $d_{i,t}^{unmet}$ in region i in period $t \in T_{test}$ by Eq. (10):

$$n_{i,t}^{unmet} = \min(\tilde{d}_{i,t}^{unmet}, d_{i,t}^{unmet}), \quad i \in I_{unmet}^{region} \quad (10)$$

where $I_{unmet}^{region} \subset I_{region}$ represents the set of the regional indices under the criteria with $\tilde{d}_{i,t}^{unmet} > \tau$ that indicates the candidates of regions potentially receive E-mopeds from additional relocation. $n_{i,t}^{unmet}$ accordingly is their maximum number of additionally relocated E-mopeds. The sum set of $n_{i,t}^{unmet}$ (sum_t^{unmet}) is subsequently utilized to obtain the unmet sum in each period $t \in T_{test}$, defined by Eq. (11):

$$sum_t^{unmet} = \left\{ \sum_{i \in I_{unmet}^{region}} n_{i,t}^{unmet} \mid t \in T_{test} \right\} \quad (11)$$

Finally, we define a set cap_t^{unmet} to describe the upper limit sum of the unmet demand, potentially satisfied by additional relocation, for each period $t \in T_{test}$ over regions. It is constrained by sum_t^{unmet} and cap_{add} , described in Eq. (12):

$$cap_t^{unmet} = \{\min(cap_{add}, sum_t^{unmet}[t]) \mid t \in T_{test}\} \quad (12)$$

Based on the above concepts with Eq. (4), Eq. (5), and from Eq. (8) to Eq. (12), we present Algorithm 1 to simulate additional relocation and its potential improvements to service capacity. Given τ , cap_{add} , $n_{i,t}^{avail}$, $d_{i,t}^{unmet}$, and $\tilde{d}_{i,t}^{unmet}$ as inputs, the algorithm initializes $\hat{n}_{i,t}^{reloc}$ as 0, which will be updated in the following procedure. We first obtain $n_{i,t}^{avail}$, cap_i^{avail} and cap_t^{unmet} , followed by computing a for loop to consider additional relocation in each $t \in T_{test}$. In each t , the negative relocation is executed before the positive relocation because the

total available number of E-mopeds for dispatch is generally significantly less than the total unmet demand in the same period in practice. $\hat{n}_{i,t}^{reloc}$ is updated by the minimum between $n_{i,t}^{avail}$ and cap_i^{unmet} until the sum of satisfaction over regions reaching cap_i^{unmet} . Noted that similar to Algorithm S1 (see Supplementary Materials 2), regions with a higher number of available E-mopeds in the same period are given priority in negative relocation. After negative relocation is assigned, cap_i^{avail} should also be decreasingly updated by $\hat{n}_{i,t}^{reloc}$ for the corresponding regions that provided E-mopeds. Next, we update the positive relocation $\hat{n}_{i,t}^{reloc}$ by $n_{i,t}^{unmet}$, with the similar principle that higher-demand regions are potentially assigned E-mopeds with a higher priority. The sum of positive relocation must be consistent with the sum of negative relocation. After the accomplishment of the additional relocation simulation, we can update $n_{i,t}^{avail}$ by the Eq. (4) and evaluate Δ_t as outputs based on Eq. (5). The above procedure is iteratively executed until the for loop concludes.

Algorithm 1. Evaluation on potential usage improvement from additional relocation

Input: τ , cap_{add} , $\tilde{d}_{i,t}^{unmet}$, $d_{i,t}^{unmet}$, $n_{i,t}^{avail}$

Initialize $\hat{n}_{i,t}^{reloc} = 0, \forall i \in I_{region}$ and $\forall t \in T_{test}$;

$cap_i^{avail} = \{\min(\{n_{i,t}^{avail} | t \in T_{test}\}) | i \in I_{region}\}$;

Obtain the set of regional indices (I_{region}^{unmet}) under the criteria with $\tilde{d}_{i,t}^{unmet} > \tau$;

$sum_t^{unmet} = \{\sum_{i \in I_{region}^{unmet}} n_{i,t}^{unmet} | t \in T_{test}\}$, where $n_{i,t}^{unmet} = \min(\tilde{d}_{i,t}^{unmet}, d_{i,t}^{unmet})$, $i \in I_{region}^{unmet}$;

$cap_t^{unmet} = \{\min(cap_{add}, sum_t^{unmet}[t]) | t \in T_{test}\}$;

for $t \in T_{test}$ **do**

Sort cap_i^{avail} in descending order with its regional sorted index set (I_{region}^{avail});

for $j \in I_{region}^{avail} \setminus I_{region}^{unmet}$ **do**

Accumulate $\hat{n}_{i=j,t}^{reloc}$ as $\widehat{acc}_t^{reloc-}$: $\widehat{acc}_t^{reloc-} += \hat{n}_{i=j,t}^{reloc}$

if $|\widehat{acc}_t^{reloc-}| \leq cap_t^{unmet}[t]$ **then**

additional negative relocation: $\hat{n}_{i=j,t}^{reloc} = -\min(n_{i=j,t}^{avail}, cap_i^{avail}[j])$

Update $cap_i^{avail}[j] = cap_i^{avail}[j] + \hat{n}_{i=j,t}^{reloc}$

else

additional negative relocation:

$\hat{n}_{i=j,t}^{reloc} = -\min(n_{i=j,t}^{avail}, cap_i^{avail}[j]) + |\widehat{acc}_t^{reloc-}| - cap_t^{unmet}[t]$

Update $cap_i^{avail}[j] = cap_i^{avail}[j] + \hat{n}_{i=j,t}^{reloc}$

break

Sort I_{region}^{unmet} as $I_{region}^{unmet, sort}$

for $k \in I_{region}^{unmet, sort}$ **do**

Accumulate $\hat{n}_{i=k,t}^{reloc}$ as $\widehat{acc}_t^{reloc+}$: $\widehat{acc}_t^{reloc+} += \hat{n}_{i=k,t}^{reloc}$

if $\widehat{acc}_t^{reloc+} \leq |\widehat{acc}_t^{reloc-}|$ **then**

additional positive relocation: $\hat{n}_{i=k,t}^{reloc} = n_{i=k,t}^{unmet}$

else

additional positive relocation: $\hat{n}_{i=k,t}^{reloc} = n_{i=k,t}^{unmet} + \widehat{acc}_t^{reloc+} - |\widehat{acc}_t^{reloc-}|$

break

$\Delta_t \equiv \sum_{i=0}^{N_{region}-1} (\hat{n}_{i,t}^{reloc}, \forall \hat{n}_{i,t}^{reloc} > 0)$

Output: Δ_t

4. Experimental results

Before evaluating relocation improvement, the demand prediction model is executed. The dataset is partitioned into training, validation, and testing parts in terms of experimental settings. The data from April 30th (representing the final day in the dataset) is designated as the test data for relocation simulation and improvement evaluation. The remaining data is then randomly divided into training and validation datasets in a 9:1 ratio. The duration of each prediction and relocation period is set to 4 hours. We utilize the previous 3-period unmet demand to predict the one in the next period. With a learning rate of 10^{-5} and a batch size of 8, the model is trained using the Adam optimizer for 2500 epochs with early stopping. The graph, node, and edge feature sizes are 8, 47, and 3, respectively. The model includes 128 convolutional hidden layers. In the training process, we fine-tune the hyperparameters of the spatio-temporal model and training process to obtain the best performance with the lowest MAE and MAPE. To ensure the model generalization with random-split data, each experiment with the same hyperparameters undergoes five repetitive training sessions to derive average evaluation metrics.

4.1. Sensitivity analysis for simulation scenarios

A sensitivity analysis is conducted to verify the feasibility and stability of the simulation for the regional E-moped situations and explore the influence of the different simulation scenario parameters, including the existing relocation capacity (cap_{exist}) and existing relocation type. Empirical observations in Greater Taipei indicate that, on average, a truck can transport an average of 25 E-mopeds within 4 hours. Presuming the existing relocation involves 5 to 10 trucks in each period, cap_{exist} ranges from 125 to 250 E-mopeds. Various potential types of existing relocation are considered, denoted from *i* to *iv*. Type *i* prioritizes executing negative relocation to regions with larger positive subtractions between drop-off and pick-up demands, while regions with smaller negative subtractions prioritize executing positive relocation. Similar to Type *i*, Type *ii* additionally introduces the Boltzmann probability distribution (Landau & Lifshitz, 2013) to add randomness and noise. Types *iii* and *iv* involve two random decisions with different random seeds to determine the relocation order. The results in Table 1a indicate that the average number of positive relocations increases with the rise in cap_{exist} . It

Table 1

Sensitivity analysis for simulation scenarios.

(a) The results of the average number of positive relocation			Existing relocation capacity (cap_{exist})					
(Unit: E-moped)			125	150	175	200	225	250
Existing relocation Type ^a	<i>i</i>		0.7	0.8	1.0	1.1	1.2	1.3
	<i>ii</i>		0.7	0.8	1.0	1.1	1.2	1.3
	<i>iii</i>		0.7	0.9	1.0	1.1	1.2	1.3
	<i>iv</i>		0.7	0.9	1.0	1.1	1.2	1.3
(b) The results of the average number of available E-mopeds								
(Unit: E-moped)			Existing relocation capacity (cap_{exist})					
			125	150	175	200	225	250
Existing relocation Type ^a	<i>i</i>		44.0	45.0	44.3	43.9	44.8	45.7
	<i>ii</i>		44.5	43.9	43.7	44.1	45.3	46.8
	<i>iii</i>		48.0	48.0	47.8	47.6	46.3	45.9
	<i>iv</i>		48.7	48.9	49.3	50.6	49.9	48.2

^a The existing relocation Type *i* prioritizes executing negative relocation to regions with larger positive subtractions between drop-off and pick-up demands, while regions with smaller negative subtractions prioritize executing positive relocation. Similar to Type *i*, Type *ii* introduces the Boltzmann probability distribution to add randomness and noise. Types *iii* and *iv* involve two random decisions with different random seeds to determine the relocation order.

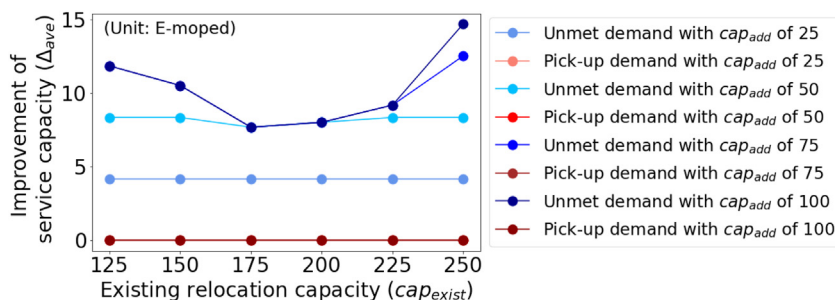


Fig. 7. Compare the potential improvement contributed by the unmet and pick-up demand-driven relocation policy. Note that cap_{add} and cap_{exist} are relocation scenario parameters, representing existing and additional relocation capacity (i.e., the maximum number of E-mopeds), respectively.

Table 2Comparison of the potential contributions between the unmet and pick-up demand-driven relocation under different τ .

Demand prediction	Threshold τ	Improvement of service capacity Δ (Unit: E-moped)							avg.
		$t = 0$	$t = 4$	$t = 8$	$t = 12$	$t = 16$	$t = 20$	total	
Unmet	0	48	0	0	0	0	0	48	8.0
	1	50	50	19	0	0	0	119	19.8
	2	50	50	50	45	0	1	196	32.7
	3	50	50	50	50	48	38	286	47.7
Pick-up	0	0	0	0	0	0	0	0	0.0
	1	50	0	0	0	0	0	50	8.3
	2	50	47	0	0	0	0	97	16.2
	3	50	50	0	0	0	0	100	16.7

Note: τ represents the setting of a crucial threshold determining positive or negative relocation, allowing E-mopeds to be relocated to other highly demanding regions when the demand in specific regions falls below this threshold. Improvement including Δ_t , Δ_{total} and Δ_{avg} respectively represent the period t , total, and average potential improvement results for the additional relocation (i.e., the number of relocated E-mopeds) based on demand predictions.

aligns with expectations because the rising relocation capacity refers to the service operator utilizing more trucks for relocation tasks. By contrast, different existing relocation types have minimal impact on the average number of positive relocations, with only a 0.1 difference when cap_{exist} is set to 150. It also meets expectations. Despite relocation types varying, the maximum number of relocated E-mopeds is primarily controlled by cap_{exist} . Different simulation algorithms all choose to place E-mopeds in those regions with unmet demand. In Table 1b, the results show that the average number of available E-mopeds remains relatively stable as cap_{exist} increases. For example, for Type *i*, the mean is 44.6 (E-mopeds), with a standard deviation of 0.67. Compared across different existing relocation types, Types *i* and *ii* exhibit similar outcomes (with mean values of 44.6 and 44.7), both relatively lower than Types *iii* and *iv* (with mean values of 47.3 and 49.3). Therefore, it can be concluded that changes in cap_{exist} have minimal impact on the available E-mopeds, while different existing relocation types have a relatively greater influence. It is reasonable since assigning E-mopeds to different regions likely affects region status in the following periods. Type *i* with fewer available E-mopeds would result in additional relocation simulations being performed more conservatively. As it is also consistent with practical relocation strategies, we select Type *i* for further experiments.

4.2. Comparison of potential contribution between unmet and pick-up demand approaches

To compare the real-time additional relocation driven by the unmet and pick-up demand, we utilize their respective predictions to evaluate potential improvement in service capacity. We define different scenarios to explore various possibilities. In the scenario where

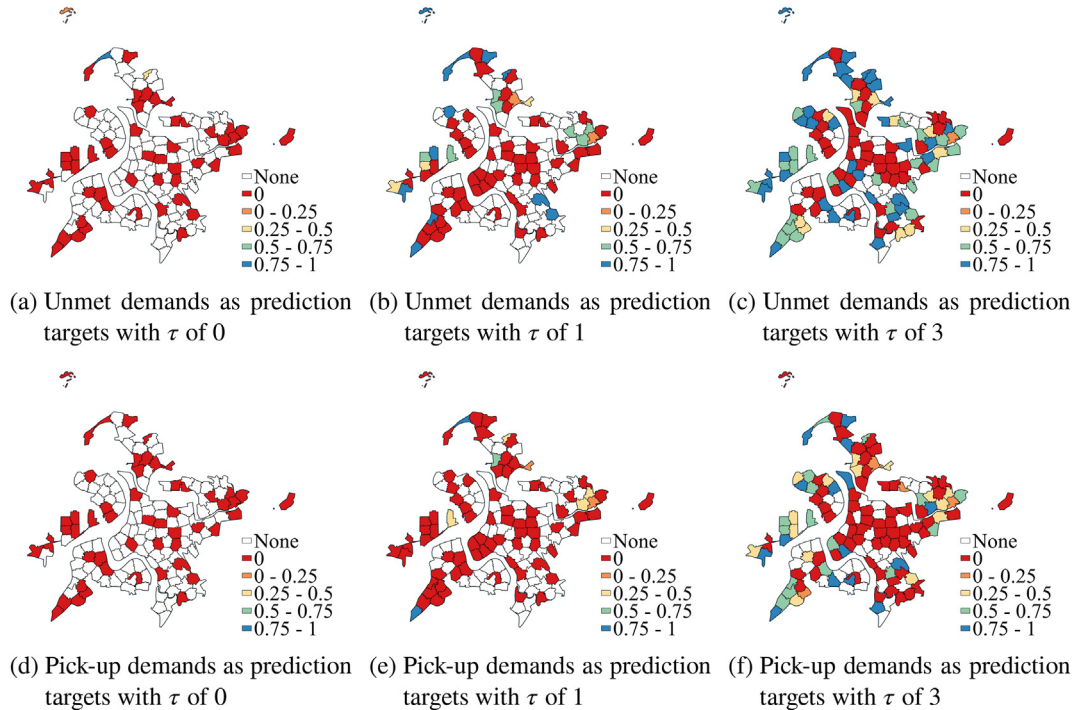


Fig. 8. Comparing the accuracy of unmet and pick-up demand predictions in each region when the ground truth is less than or equal to τ .

the relocation threshold parameter (τ) is fixed at 0 (E-moped) and existing relocation is set as Type i , we establish various combinations of the existing relocation capacity (cap_{exist}) and additional relocation capacity (cap_{add}). The results of average potential improvement (Δ_{avg}) (E-moped/period) are demonstrated in Fig. 7. The noteworthy observation is the significantly superior performance of the unmet demand-driven relocation policy over the pick-up demand-driven policy. The four lines representing pick-up demand with cap_{add} from 25 to 100 (E-mopeds) overlap with each other, indicating that regardless of the varying cap_{exist} and cap_{add} , the improvement in relocation facilitated by the pick-up demands remains consistently at 0. However, the unmet demand-driven approach exhibits more favorable outcomes. For cap_{add} of 25 and 50 (E-mopeds), the average improvements remain almost constant at 4.2 and 8.2, respectively, regardless of changes in cap_{exist} . Concerning cap_{add} of 75, the average improvement varies with changes in cap_{exist} starting at 11.8, reaching a minimum of 7.7, and rising to 12.5. The results for cap_{add} of 100 are similar to those of 75 but rise to 14.7 when cap_{exist} is 250 (E-mopeds). Generally, the increase in cap_{add} contributes to the service capacity; however, corresponding to different cap_{exist} there are upper bounds for the improvements, such as 7.7, 8, and 9.2 (E-mopeds/period) when cap_{exist} is 175, 200, and 225, respectively.

We tend to explore the impact of τ on relocation. By fixing the existing relocation capacity (cap_{exist}) at 200 (E-mopeds), additional relocation capacity (cap_{add}) at 50 (E-mopeds), and employing existing relocation Type i , we evaluate the changes in improvement of service capacity Δ_t corresponding to different relocation threshold (τ) in Table 2. In setting τ to 0, the unmet demand-driven Δ_t is 48, 0, 0, 0, 0, and 0 (E-moped), while the pick-up demand-driven Δ_t is 0, 0, 0, 0, 0, and 0, corresponding to each period, respectively. As τ increases to 1, the unmet demand-driven Δ_t enhances to 50, 50, 19, 0, 0, and 0 (E-moped), while the pick-up demand-driven Δ_t enhances to 50, 0, 0, 0, 0, 0. Setting τ to 2 results in the unmet demand-driven Δ_t of 50, 50, 50, 45, 0, and 1 (E-moped), and pick-up demand-driven Δ_t of 50, 47, 0, 0, 0, and 0. When setting τ to 3, the unmet demand-driven results demonstrate Δ_t of 50, 50, 50, 50, 48, and 38 (E-moped), while the pick-up demand-driven results show Δ_t of 50, 50, 0, 0, 0, and 8 corresponding to each period. Several key observations should be noted when comparing unmet and pick-up demand results. First, with the increase in τ , Δ increases, and the number of affected periods also increases accordingly, particularly for the unmet demand-driven policy in setting τ to 3. Second, from the perspective of Δ_{total} , differences between the unmet and pick-up demand under τ of 0, 1, 2, and 3 are 48, 69, 99, and 186 (E-mopeds), respectively. It indicates that with the increase in τ , the unmet demand-driven policy increasingly contributes more to service capacity than pick-up demand. Third, potential improvement is relatively challenging in the last two periods, possibly due to their highest average demand shown in Fig. 6b. In general, regardless of changes in τ , the unmet demand-driven results consistently outperform pick-up demand in both Δ_{total} and Δ_t in each period.

4.3. Comparison in scenario of low demands between unmet and pick-up approaches

To understand the reasons for the superiority of the unmet demand-driven policy over pick-up demand in the results, we compare them with the ground truth of unmet demand separately in terms of spatial distribution. As the critical decision for positive or negative relocation is based on the threshold τ , the prediction accuracy corresponding to its value becomes crucial. We examine the performance

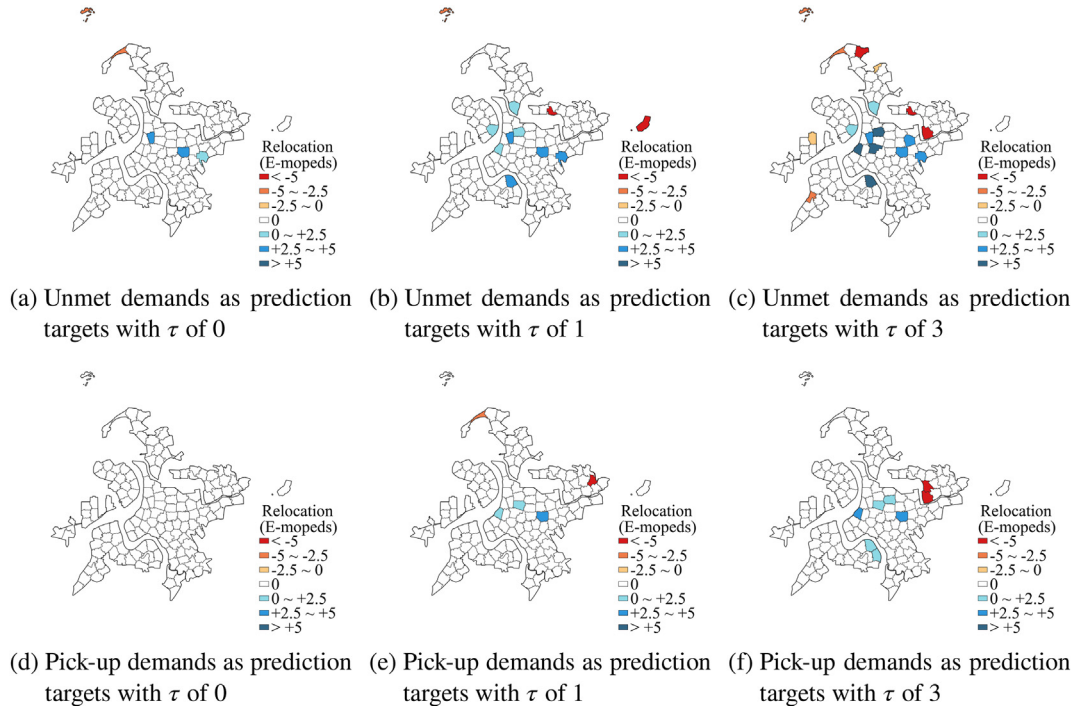


Fig. 9. Comparing the average number of additional relocations based on unmet and pick-up demand predictions in each region under different τ .

for the ground truth less than or equal to τ and evaluate the accuracy of unmet and pick-up demand separately in this context. With the existing relocation capacity (cap_{exist}) fixed at 200 (E-mopeds), the additional relocation capacity (cap_{add}) at 50 (E-mopeds), and defining existing relocation Type i , we demonstrate the accuracy of the unmet and pick-up demand in each region throughout a day under different τ in Fig. 8. White regions indicate no ground truth less than or equal to τ in those areas, while red, yellow, green, and blue regions indicate increasing ranges of ground truth accuracy intervals.

Compared to the results of unmet demand prediction between Fig. 8a and d, accuracy in each region under the relocation threshold (τ) of 0 is challenging for both. The pick-up prediction fails to accurately predict regions with a demand of 0, while the unmet prediction can identify three regions with a demand of 0 in the north, with accuracy of 0.17, 0.25, and 1.00, respectively. Correct predictions for these three regions all occur from 0 a.m. to 4 a.m., as shown in the result of 48 in Table 2. When setting τ to 1, the ability to accurately predict the ground truth less than or equal to 1 improves in various regions. Comparing the results of unmet and pick-up demand, 3 and 4 regions with accuracy greater than or equal to 0.25, 7 and 1 regions with accuracy greater than or equal to 0.50, and 12 and 2 regions with accuracy greater than or equal to 0.75, respectively. With τ set to 3, the ability to accurately predict the ground truth less than or equal to 1 significantly improves in more regions. The unmet and pick-up demand results obtain 10 and 15 regions with accuracy greater than or equal to 0.25, 23 and 16 regions with accuracy greater than or equal to 0.50, and 35 and 15 regions with accuracy greater than or equal to 0.75. Overall, the specified performance differences between the unmet and pick-up demand approach become more pronounced with an increase in τ . It not only indicates the importance of τ can mitigate the negative impact of inaccurate predictions but also explains why the unmet demand-driven relocation policy significantly outperforms pick-up demand in enhancing service capacity.

5. Discussion

In Section 4.2 and 4.3, we have demonstrated the advantages of the unmet demand-driven relocation rather than the pick-up demand-driven. Nevertheless, we can further delve into the additional relocation situation in each region in Fig. 9 from an example of a scenario with fixing cap_{exist} at 200, cap_{add} at 50, and existing relocation Type i . The red color denotes the negative relocation, with darker shades indicating a larger average number; The blue color denotes the positive relocation, with darker shades indicating a larger average number. Compared with Fig. 9a, b, and 9c, it can be seen that in setting τ to 0, only the three central regions (around Taipei Main Station, Eastern District, and Xinyi) can receive more E-mopeds from the relocation, originating from the two regions in the northwest (around Danshui and Beitou). When setting τ to 1, additional regions' demand can be fulfilled, and the idle E-mopeds also originate from two additional regions (around Dazhi and Xizhi). When setting τ to 2, more regions, such as around Xinzhuang and Far Eastern Memorial Hospital, can provide available E-mopeds. Compared with the pick-up results, with the same τ , the regions where the unmet demand-driven relocation policy executes positive relocation overlap with the pick-up and are more numerous. It indicates that both approaches have similar predictions for regions with the highest demand, yet the relocation performance of the unmet demand is much superior. Precisely, corresponding to τ of 0, 1, and 3, the numbers of the identified regions for the positive relocation are 3 and 0, 8 and 3, and 10 and 6 for the unmet and pick-up demand, respectively. These results of the identified high-demand regions are consistent with the ground truth analysis in Fig. 5c. Generally, the unmet demand-driven relocation can assist in more high-demand regions and the more suggested regions for negative relocation. It also indicates the advantage of flexible route choices for transportation trucks. Each truck can be assigned tasks with shorter multiple-region routes for pick-up and drop-off, thus reducing total delivery time. It assists in avoiding discrepancies between the supply and demand times, reducing misses, and improving the fulfillment of potential unmet demands.

This study aims to reduce unmet demand through service providers' real-time relocation. In practice, service operators implement daily relocation strategies focusing on similar regions based on long-term historical experience. Instead, we propose in-time relocation policies with dynamic relocation regions. Our proposal can more effectively fulfill in-time needs in high-demand regions with limited relocation capacity. As an increase in relocation capacity, the relocation coverage of regions with unmet demand could also be improved. Based on scenarios with different relocation capacities in our research, Service operators could evaluate a mutually beneficial scheme between the relocation costs (such as trucks and workers) and the enhancement of customer revenue.

From a broader perspective of urban sustainable development, the government should improve relevant policies to promote the development of shared E-moped services. Following relevant regulations, the total number of shared E-mopeds permitted in Taipei City was rarely changed and limited to about 11,400 in 2020, 13,300 in 2022, and 12,800 in 2024, respectively (Department of Transportation, Taipei City Government, 2024). In response to the continuous increase in unmet demand, the government may consider appropriately relaxing the restriction on the maximum size of the E-moped fleet. In addition, according to current regulations, a single service operator cannot place more than three shared E-mopeds consecutively, and no more than six are allowed in the same block (Department of Transportation, Taipei City Government, 2023). This regulation possibly leads to challenges in relocation and difficulties in satisfying user demand in some high-demand regions. Based on our research results, some high-demand regions can be identified in specific periods, especially the regions where unmet demand cannot be fulfilled even after relocation. Accordingly, the government could consider more flexible regulatory frameworks for shared E-moped parking to adapt to different urban areas and times of demand. In addition, establishing designated parking zones for shared E-mopeds is also a feasible strategy to facilitate efficient relocation and accessibility.

Our policy is adaptive to other cities. Scaling our model to larger cities involves retraining it with specific unmet demand and trip data from those locales based on the appropriate division of the target regions. Because an increase in the number of regions may result in a slight increase in error, the relocation threshold (τ) setting also needs to be adjusted accordingly to obtain a better relocation policy with improved service capacity. When applied to more complex urban environments, the model would require more relevant built environment data, including more intricate public transportation systems (such as ferries and shuttle buses). Although expanding the model increases the complexity and the data required, our graph-based approach with enhanced node features is designed to

accommodate this increase efficiently. While the training time for the model may extend as it scales, the method still supports the derivation of pre-trained models capable of delivering real-time predictions of unmet demand. Our unmet demand-driven policy can also be implemented in other modes of shared micro-mobilities, such as shared bikes and E-scooters, which are also considered the first-and-last-mile connection to public transportation. They also rely on similar service relocating operations with delivery trucks. The loading capacity of trucks should be reconsidered in terms of consistency with their smaller unit volumes. As bikes and scooters are often used for shorter distances, the importance of built environment features likely changes and needs to be retrained. The Periodic and weather conditions still play relatively essential roles. Neither can provide a canopy for users, and citizens' daily activities deeply influence their user behavior. By contrast, the shared vehicle is not suggested to use our approach because of its long-distance travel patterns, canopy providing, and different service relocation management.

On the other hand, to discuss the limitations of our research, this research primarily focuses on elucidating the importance of the unmet demand-driven relocation policy and the caution against replacing it with the pick-up, as seen in previous studies. Therefore, we did not discuss the optimization of relocation routing in terms of time, distance, and material costs. In our current model, battery management is simplified based on the operational strategies employed by service operators. Specifically, our assumptions are grounded in the prevalent practice of battery swapping, where workers ensure a continuous supply of fully charged batteries at dispatch stations. This setup minimizes the impact of battery charging times and the availability of charging infrastructure on our simulation results. As such, our model presumes that E-mopeds always possess sufficient battery power for their intended rental tasks, which sidesteps the complexities related to real-time battery level monitoring and charging logistics. However, we acknowledge that exploring the dynamics of real-time battery levels and their effects on user behavior and unmet demand could be a valuable extension of this research in the future. Another limitation of our study is the equitable accessibility of shared scooters for different community groups. We primarily focus on optimizing service capacity based on available historical data. Unfortunately, current services are currently concentrated in urban and densely populated areas. To effectively address this gap, it is necessary to undertake holistic planning that incorporates the development of communities, enhances the integration of E-moped services with existing public transportation networks, and analyzes specific transportation needs and usage demands of under-served areas. With government support and subsidies, this service could better meet the diverse needs of all community segments (Aman et al., 2021; Brown, 2021).

In future research, with comprehensive data on regional E-moped situations and unmet demand information, we plan to leverage the findings of this study to explore the optimization of complex relocation routing. With centralized training and decentralized execution in deep reinforcement learning methods, each delivery truck is considered an independent actor that proposes which regions to pass through for pick-up and drop-off actions. A joint linear program method is then used to connect these regions into the shortest paths, and each is calculated for its reward, average loading, path cost, and other information. Subsequently, a graph neural network model is employed to establish relationships between the trucks, passing their rewards and the information above to each other. The critic networks are built to consider individual actors' optimization and their integrated optimization comprehensively. Once this model is trained, inputting the unmet demand prediction for each region and relevant environmental information into the trained actors can propose an optimal real-time relocation routing strategy. This method will include planning multi-region dispatch paths and simultaneous use of trucks with different capacities. The total fleet of trucks and E-mopeds should also be considered as a constraint of service operating cost. The optimization goal is to improve the best service capacity while bearing a specific allowable operating cost. Its multi-objective problems can be addressed by applying the Pareto frontier (He & Shin, 2020).

6. Conclusion

This research emphasizes unmet demand's crucial and indispensable role in the relocation policy, which has been omitted in previous studies. To substantiate this statement, we propose a research framework to evaluate the potential improvement, comparing the unmet demand-driven policy with the pick-up demand-driven policy. We propose a spatio-temporal simulation of regional E-moped situations based on historical trip data, adjustable by relocation parameters for various scenarios. A linear program evaluation method evaluates potential capacity improvements in re-utilizable idle E-mopeds resulting from additional relocations. Our findings consistently demonstrate the superiority of unmet demand-driven relocation over pick-up demand in every simulation scenario. Specifically, with the relocation threshold (τ) of 0 (E-moped), daily total potential improvements in service capacity are 48 and 0 (E-moped) for unmet and pick-up demand, respectively. Here, τ determines that if the demand in a region falls below the designated threshold, its E-mopeds should be dispatched to other regions. As τ increases to 3, the daily total potential improvement rises to 286 and 100 (E-mopeds) for the unmet and pick-up demand, respectively. These differences are evident not only in total potential improvements but also in regional results. Unmet demand-driven relocation accurately identifies more regions with idle E-mopeds for relocation to high-demand regions. With τ of 0, 1, and 3, the identified regions for positive relocation execution are 3 and 0, 8 and 3, and 10 and 6 (E-mopeds) for the unmet and pick-up demand, respectively.

We highlight that our research could contribute to various aspects. Concerning city sustainable development and the reduction of private motorcycles, city governments should concentrate on developing shared E-moped and improve related regulations based on the usage and unmet demand situations. Shared E-moped parking spaces should be ensured to facilitate efficient relocation and accessibility. On the other hand, our policy could be implemented in cities with more extensive scales and complex transportation systems. Shared bikes and E-scooters are also suitable application domains for our research. Besides considering the different built environment factors, our model can easily convert them as graph-representation features. Note that the importance of different features may change in different cities with different types of vehicles after retraining the prediction model. The relocation threshold also needs to be reconsidered according to the simulation results and reflection of service relocation capacity.

While this study significantly clarifies the importance of addressing unmet user demand in relocating shared E-mopeds, it is also

essential to acknowledge some limitations. First, the proposed evaluation of potential improvement calculates the potential upper bound without accounting for certain constraints, such as the total fleet of delivery trucks and E-mopeds and the dispatch cost. Our future research will acquire comprehensive data on regional E-moped situations and unmet demand information to explore real-time relocation routing strategies. Leveraging the graph reinforcement learning model, we will consider multiple trucks with different capacities to execute individual relocation jobs with multi-point backhauls, addressing multi-objective problems involving maximizing unmet demand satisfaction and minimizing operating costs. Integration of traffic conditions and relocation strategies at each period is also worth studying to consider relocation time cost and truck routing better. Severe traffic congestion likely reduces the actual effectiveness of relocation, so alternative routing plans should be proposed accordingly. On the other hand, we suggest that future research could focus on improving the accuracy of the unmet demand prediction model, as providing accurate spatial-temporal predictions for every small segmented region remains challenging. Precise prediction for values close to zero warrants attention, as it significantly influences identifying regions where idle E-mopeds should be allocated, particularly in scenarios where the total demand significantly exceeds the supply.

CRedit authorship contribution statement

Jia-Cherng Song: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **I-Yun Lisa Hsieh:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Chuin-Shan Chen:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jum.2025.04.001>.

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